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EXP.NO: 4 QUERYING LEDGER DATA WITH HYPERLEDGER FABRIC

AIM

To query and retrieve ledger data from a Hyperledger Fabric blockchain network using JavaScript chaincode, and verify asset information stored in the world state database.

PROCEDURE

1. Navigate to the Hyperledger Fabric test network directory:

```
cd ~/fabric-dev/fabric-samples/test-network
```

2. Start the Fabric network and create the application channel:

```
./network.sh up createChannel -ca
```

3. Deploy the JavaScript Asset Transfer chaincode:

```
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/
chaincode-javascript
```

4. Initialize the ledger with sample asset data:

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride
orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c
'{"function":"InitLedger","Args":[]}'
```

5. Query all asset records stored in the ledger:

```
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

6. Query specific asset record (asset1):

```
peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset1"]}'
```

7. Create a new asset (asset20) and query state:

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride
orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c
'{"function":"CreateAsset","Args":["asset20","Green","8","Arun","1500"]}'
```

8. Query non-existing asset (asset50) to verify error handling:

```
peer chaincode query -C mychannel -n basic -c '{"Args":
["ReadAsset","asset50"]}'
```

9. Stop the Fabric network:

```
./network.sh down
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -ca
Creating channel 'mychannel'... done
Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -ccl javascript -ccn
basic -ccp ../asset-transfer-basic/chaincode-javascript
Chaincode committed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --
ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic
-c '{"function":"InitLedger","Args":[]}'
Ledger Initialized Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic
-c '{"Args":["ReadAsset","asset1"]}'
{
  "ID": "asset1",
  "Color": "blue",
  "Size": 5,
  "Owner": "Tomoko",
  "AppraisedValue": 300
}
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --
ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic
-c '{"function":"CreateAsset","Args":["asset20","Green","8","Arun","1500"]}'
Asset asset20 created successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic
-c '{"Args":["ReadAsset","asset20"]}'
{
  "ID": "asset20",
  "Color": "Green",
  "Size": 8,
  "Owner": "Arun",
  "AppraisedValue": 1500
}
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic
-c '{"Args":["ReadAsset","asset50"]}'
Error: The asset asset50 does not exist.
```

RESULT

The ledger data stored in the Hyperledger Fabric blockchain network was successfully queried using JavaScript chaincode. Asset records were retrieved from the world state database, specific asset details were verified, and newly created assets were successfully queried, demonstrating efficient data retrieval and verification in a permissioned blockchain environment.